MIL-P-50944B (1R)
26 May 1982
SUPERSEDING
MIL-P-50944A (PA)
21 July 1975

MILITARY SPECIFICATION

PROJECTILE, 155MM, HE: M483A1
NON-METALLIC HARDWARE FOR

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, provides quality assurance provisions, and packaging for components designated as non-metallic hardware, for use with Projectile, 155MM, HE: M483A1.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards - Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

FSC 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.2 Other Government documents, drawings, and publications - The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

9215203 - Hardware, non-metallic

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence - In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts - The parts shall comply with all requirements specified on Drawing (Dwg) 9215203, all associated drawings, and with all requirements specified in applicable specifications.

3.3 First article testing - This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

3.4 Workmanship - All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All marking and stamping shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections - The following types of inspection shall be conducted on this item:

a. First article inspection
b. Quality conformance inspection

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4.3 First article inspection

4.3.1 Submission - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Forward Plate Pads	9216853	25
Spacers	9216846-1	25
Spacers	9216846-2	25
Sleeves	9216850-1	25
Sleeves	9216850-2	25

4.3.2 Inspections to be performed - See MIL-A-48078 and Table 1 specified herein.

4.3.3 Rejection - See MIL-A-48078.

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TABLE I - First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50944B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Non-Metallic Hardware				A-9215203
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Forward plate pads (9216853) Examination for defects	25	—	3.2	4.4.2.1
	Spacers (9216846-1) Examination for defects	25	—	3.2	4.4.2.2
	Spacers (9216846-2) Examination for defects	25	—	3.2	4.4.2.2
	Sleeves (9216850-1) Examination for defects	25	—	3.2	4.4.2.3
	Sleeves (9216850-2) Examination for defects	25	—	3.2	4.4.2.3

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation - Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination - See MIL-A-48078.

a. Sampling plans - Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50944B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Pad, forward plate				9216853 NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter		0.40%	3.2	Gage
102	Outside diameter, max.		0.40%	3.2	Gage
Minor					
201	Thickness		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50944B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9216846-1 9216846-2
4.4.2.2	Spacer				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Total width		0.40%	3.2	Gage
102	Total height, max.		0.40%	3.2	Gage
103	Distance to basic width on largest radius surface		0.40%	3.2	Gage
104	Distance to widest basic width between small radii surfaces		0.40%	3.2	Gage
105	Distance to smaller basic width between small radii surfaces		0.40%	3.2	Gage
106	Symmetry of basic width on largest radius surface		0.40%	3.2	Gage
107	Symmetry of widest basic width between small radii surfaces		0.40%	3.2	Gage
108	Symmetry of smaller basic width between small radii surfaces		0.40%	3.2	Gage
109	Length, max.		0.40%	3.2	Gage
110	Surface variation on large radius surface in excess of permitted depth (not to include gating area)		0.40%	3.2	Gage
111	Surface variations on large radius surface outside of permitted zone (4 places)		0.40%	3.2	Gage

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50944B (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9216846-1 9216846-2
4.4.2.2	Spacer (cont.)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major (cont)					
112	Gating area, ejection pin marks, mold identification above flush		0.40%	3.2	Visual/Manual
113	Height of rail from surface (2 places)		0.40%	3.2	Gage
114	Width of rail, min. (2 places)		0.40%	3.2	Gage
Minor					
201	Length, min.		0.65%	3.2	Gage
202	Surface variation on top surface in excess of specified limit		0.65%	3.2	Gage
203	Gating area in excess of specified length or width		0.65%	3.2	Gage
204	Depth of gating area excessive (as measured from surface around gating area)		0.65%	3.2	Gage
205	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50944B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9216850-1 9216850-2
4.4.2.3	Sleeve				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness		0.40%	3.2	Gage
102	Height		0.40%	3.2	Gage
103	Inside radius		0.40%	3.2	Gage
104	Length, max.		0.40%	3.2	Gage
105	Surface variation of outside surface in excess of permitted depth (not to include gating area)		0.40%	3.2	Gage
106	Surface variation on outside surface outside of permitted zone (2 places)		0.40%	3.2	Gage
107	Gating area, ejection pin marks, or mold identification above flush		0.40%	3.2	Visual/Manual
108	Height of ribs (2 places)		0.40%	3.2	Gage
109	Location of ribs (2 places)		0.40%	3.2	Gage
Minor					
201	Length, min.		0.65%	3.2	Gage
202	Gating area in excess of specified length or width		0.65%	3.2	Gage
203	Depth of gating area excessive (as measured from surface around gating area)		0.65%	3.2	Gage
204	Evidence of poor workmanship		0.65%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50944B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Container, Sealed				See Section 5
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Container damaged		0.40%	3.2	Visual
102	Contents of container loose		0.40%	3.2	Visual/Manual
103	Container incorrect according to size and weight		0.40%	3.2	Visual
104	Closing of container, incorrect or incomplete		0.40%	3.2	Visual
105	Marking data missing, illegible		0.40%	3.2	Visual
Minor					
201	Size of marking letters incorrect		0.65%	3.2	Visual

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4.4.3 Testing

4.4.3.1 Dimensional control of stamped parts - In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts may be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.4.3.2 Dimensional control of plastic parts - In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, and after a curing time for the parts has been approved (See Note 1), a sample of three (3) parts (as molded) from each cavity may be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of each cavity during production.

The inspection of the parts shall be performed only when the curing time for the parts has been reached. The inspection for control of each cavity during production shall be performed on the above quantities from each cavity after production of each 5000 parts or at the end of each shift, providing the number of parts produced during the shift is less than 5000 parts. Of the three (3) samples, one (1) sample shall be the last part produced. Parts that fail to comply with dimensional requirements after dimensional stability has been attained shall be classed defective.

If any defective parts are found during qualification of the cavity, the cavity producing the defective parts will not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for corrective action.

All cavities formerly removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test as above. The contractor may request a change of inspection frequency providing he presents objective evidence to substantiate the request to the Contracting Officer.

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The contractor shall furnish evidence to the Government inspector that dimensional stability has been attained prior to measuring the parts for acceptance.

(Note 1) - Establishing a curing time as follows. Dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60, etc. mins.) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to Commander, US Army ARRADCOM, ATTN: DRDAR-QAR-Q, Dover, NJ 07801. If any new change in material, cycle time, a reworked cavity, or new cavity is used, a new curing time shall be established and approved.

4.4.4 Inspection equipment - The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures - This section is not applicable to this specification.

5. PACKAGING

5.1 Preservation

5.1.1 Level C - The non-metallic hardware shall be preserved and packaged in accordance with MIL-STD-1169. The items shall be bulk-packaged.

5.2 Packing

5.2.1 Level C - The non-metallic hardware, preserved and packaged as specified in 5.1, shall be packed for shipment in accordnace with MIL-STD-1169. Each pack shall contain only one part number.

5.3 Marking - Marking of the carton shall comply with MIL-STD-1169.

5.4 Shipping - When components from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use - These components are intended for use with the projectile, 155MM, HE: M483A1.

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6.2 Ordering data - See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval - See MIL-A-48078. Submit equipment designs, as required, to Commander, US Army ARRADCOM, ATTN: DRDAR-QAR-I, Dover, NJ 07801.

6.4 Drawings - Drawings listed in Section 2 of this specification under the headings U.S. Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.5 Changes from previous issue - The margins of this specification are marked with vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-50944B (AR) Projectile, 155MM, HE: M483A1 Non-Metallic Hardware For

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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